Calibration Board -- Ground Truth Measurement Data

Status: LIVE reference for this app. Measurement of record for the physical board.
Measured: 2026-08-18 by Derek, digital calipers.
In force: src/board.py uses these values as of 2026-08-18 (DEFAULT_CELL_WIDTH_MM = DEFAULT_CELL_HEIGHT_MM = 76.39 mm).
Source: references/BoardTrueSize.md. Superseded history: references/CalibrationBoard.txt.

1. The board

12 x 8 squares, 11 x 7 interior corners (77 corners total), printed on white stock mounted on cardboard, with four square-in-square fiducial markers in the corner cells.
[measured] -- confirmed by running cv2.findChessboardCornersSB over H:\Camera Calibration Files\Pave\TDR054S-C_234500499_260811190327_image24.png: locks on at 11 x 7 and returns 77 corners.

The corner fiducials are plain square-in-square marks, not ArUco patterns, and are not used for detection. Detection is plain chessboard; the ArUco module is never called (see src/detector.py).

2. Measurement of record

Quantity	Value	How
Cell pitch	76.39 mm	152.78 mm across 2 cells / 2 [measured]
Cell pitch, inches	3.0075 in	= 76.39 / 25.4 [measured]
Single square, own width	76.59 mm	calipers, edge to edge [measured]
Ink spread	0.10 mm/edge	(76.59 - 76.39) / 2 [inference]

Cross-check, steel tape over 10 cells: 765 mm -> 76.50 mm/cell [measured], 0.14% above the caliper figure. Consistent within tape resolution.

Method -- why 2 cells and not 1: measure the same edge on two cells two columns apart (white->grey to white->grey). Both jaws then sit on the same kind of transition and printed ink spread cancels. Measuring a single square's own width puts one jaw on a white->grey edge and the other on grey->white, so the spread is added to the reading -- the single square read 0.20 mm larger than the true pitch, every time, in one direction: a repeatable bias, not noise.

Anyone re-measuring this board: span 2 cells, same edge to same edge. Do not measure one square.

3. What the app uses, and why the change was safe

src/board.py sets DEFAULT_CELL_WIDTH_MM = DEFAULT_CELL_HEIGHT_MM = 76.39 [measured] as of 2026-08-18. It previously held 76.2 (3.000 in), the assumed nominal, never measured.

A uniform scale error in the board has no effect on the intrinsics [measured] -- tested directly by calibrating one fixed set of synthetic images while varying only the cell size given to calibrate().

Cell size entered	fx/fy	RMS
76.2 x 76.2 (truth)	0.999986	0.046367
77.2 x 77.2 (+1.3%)	0.999986	0.046367
70.0 x 70.0 (-8.1%)	0.999986	0.046367

3.0 x 3.0 (inches, not mm)	0.999986	0.046367
76.465 x 76.002 (non-square)	0.994037	0.046376
76.2 x 74.0 (3% off in y)	0.971850	0.046378

The object points scale, the recovered translation vectors scale with them, and fx, fy, cx, cy and the distortion coefficients come out identical -- even for a 25x error. Only the ratio between cell width and cell height moves the result.

So 76.39 and 76.2 produce identical intrinsics. The change was made because the app should state what was measured rather than what was assumed -- not because it improves anything numerically. The two calibrations already in the store were computed with 76.2 and remain valid; new calibrations record 76.39.

4. Squareness -- the thing that actually matters

The Pave calibration of record reports fx/fy = 0.999836 -- 0.016% from square [measured], across 53 images and 77 corners each. If the board were even 0.2% non-square, calibrating with square cells would have dragged fx/fy to about 0.998. It did not.

The camera measures the board's squareness roughly 10x finer than calipers can. Hand measurement is a gross-error check here (wrong board, inches/mm mix-up, a warped or shrunken board), not a refinement.

5. Open items

1. Vertical 2-cell span not yet independently measured -- cell HEIGHT = 76.39 assumes the vertical matches the horizontal. The calibration's own fx/fy = 0.999836 says the two directions agree to 0.016%, far finer than calipers resolve. Worth one reading if the board is ever reprinted. Expect ~152.78 mm.

2. Flatness not checked -- a bowed board is a real error source and no cell measurement detects it. Lay a straightedge across both diagonals.

6. When the absolute size DOES matter

Anything that solves distance, height or scale from the board's pose (rvecs/tvecs) rather than from a physical reference. Using 76.2 where the truth is 76.39 biases such a result by +0.25% [inference].

H:\Camera Calibration Files\barcheck.py measures a known-length bar, not the board, so it is unaffected today. If a board-pose method is ever added, use 76.39 mm and re-measure both directions first.